CO4 – ASSESSMENT TOOL 1

CODING CHALLENGE – CLOUD RELEASE BUDGET PLANNER

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#include <stdio.h>
#include <stdlib.h>
#define MAXN 205
#define MAXW 10005
typedef struct {
    char name[1001];
    int cost, value, index;
} Feature;
Feature f[MAXN];
static int dp[MAXN][MAXW];
int selected[MAXN];
int compare(const void *a, const void *b) {
    Feature *x = (Feature *)a;
    Feature *y = (Feature *)b;
    long long p = (long long)x->value * y->cost;
    long long q = (long long)y->value * x->cost;
    if (p > q) return -1;
    if (p < q) return 1;
    return x->index - y->index;
}
int main() {
    int n, W, i, w;
    int greedyValue = 0, greedyUsed = 0;
    scanf("%d %d", &n, &W);
    Feature sorted[MAXN];
    for (i = 1; i <= n; i++) {
```

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    scanf("%1000s %d %d",
          f[i].name, &f[i].cost, &f[i].value);
    f[i].index = i;
    sorted[i] = f[i];
}
qsort(sorted + 1, n, sizeof(Feature), compare);
for (i = 1; i <= n; i++) {
    if (greedyUsed + sorted[i].cost <= W) {
        greedyUsed += sorted[i].cost;
        greedyValue += sorted[i].value;
    }
}
for (i = 1; i <= n; i++) {
    for (w = 0; w <= W; w++) {
        dp[i][w] = dp[i-1][w];
        if (f[i].cost <= w) {
            int take = f[i].value +
                    dp[i-1][w-f[i].cost];
            if (take > dp[i][w])
                dp[i][w] = take;
        }
    }
}
int maxValue = dp[n][W];
int budget = 0;
while (dp[n][budget] != maxValue)
    budget++;
int used = budget;
int remaining = budget;
for (i = n; i >= 1; i--) {
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        if (dp[i][remaining] != dp[i-1][remaining]) {
            selected[i] = 1;
            remaining -= f[i].cost;
        }
    }
    printf("Greedy Value = %d\n", greedyValue);
    printf("Maximum Value = %d\n", maxValue);
    printf("Selected =");
    int found = 0;
    for (i = 1; i <= n; i++) {
        if (selected[i]) {
            printf(" %s", f[i].name);
            found = 1;
        }
    }
    if (!found)
        printf(" NONE");
    printf("\nBudget Used = %d\n", used);
    return 0;
}
```

Sample Input:

3 50

Auth 10 60

Search 20 100

Payments 30 120

Sample Output:

Greedy Value = 160

Maximum Value = 220

Selected = Search Payments

Budget Used = 50